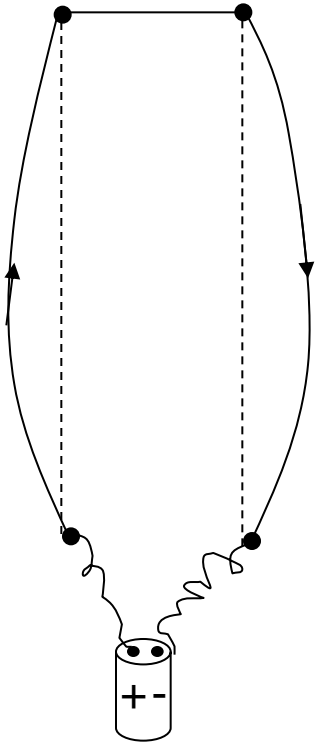
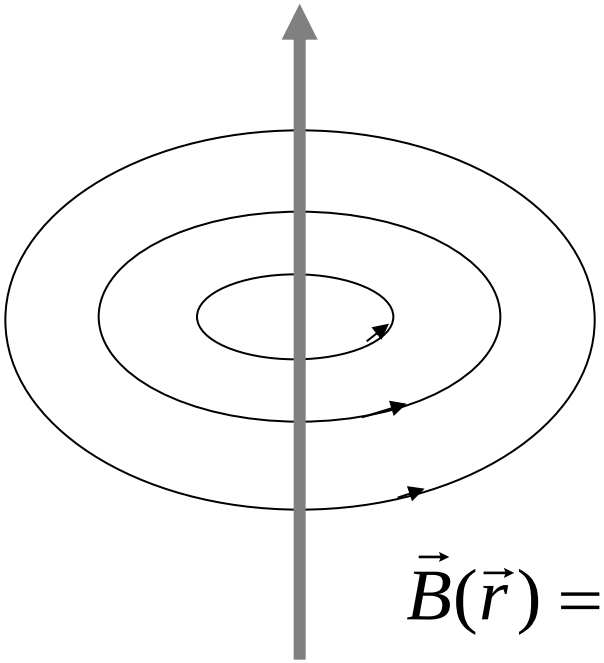
Magnetostatics



Two parallel current carrying conductors attract each other when the currents are in the same direction and repel each other when the currents are in opposite directions.



When a current flows through a straight conductor, magnetic field is generated circumferentially

$$\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} I \int \frac{d\vec{l} \times \hat{R}}{R^2}$$

Biot-Savart Law - we will see this soon

Stationary charge produces only an electric field around,
But a moving charge generates a magnetic field also.

Stationary charges --- constant electric fields: electrostatics
Steady currents --- constant magnetic fields: magnetostatics

Magnetostatics

Magnetic forces

Magnetic forces in a charge Q, moving with a velocity **v** in a magnetic field is

$$\vec{F}_{mag} = Q\left(\vec{v} \times \vec{B}\right) \longrightarrow \quad \textbf{(1)}$$

Known as Lorentz force law

In the presence of both $\vec{\mathbf{E}}$ and $\vec{\mathbf{B}}$, the net force on Q would be

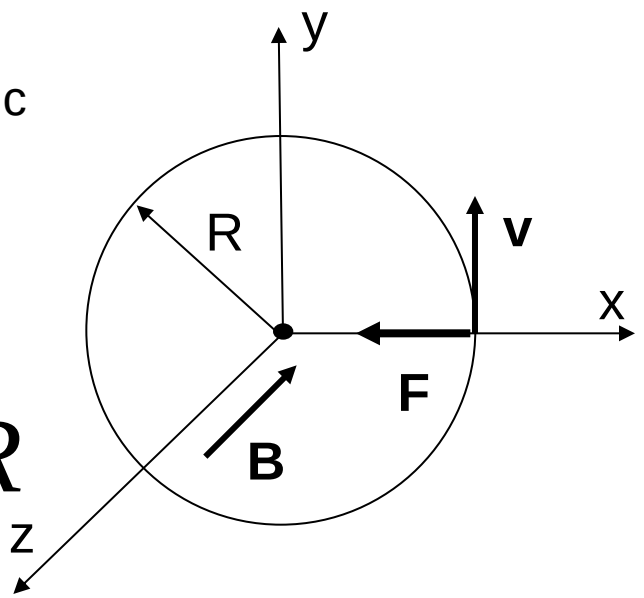
$$\vec{F} = Q\left[\vec{E} + \left(\vec{v} \times \vec{B}\right)\right]$$

Equation (1) is a fundamental axiom of the theory, whose justification is found in experiments

Example

Cyclotron motion

The motion of a charged particle in a magnetic field is circular.



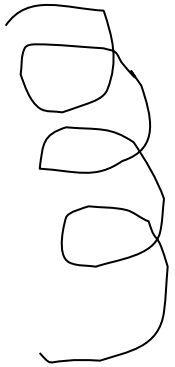
Magnetic force = Centripetal acceleration

$$QvB = m\frac{v^2}{R} \quad \text{or} \quad p = QBR$$

Q – charge, v- speed, m – mass, p is the momentum of the particle

This is Cyclotron formula

If the particle has an additional speed $v_{||}$ to B, then the particle will moves in a helix



Magnetostatics

Magnetic forces do no work

We have seen that the Magnetic force $\vec{F}_{mag} = Q\left(\vec{v} \times \vec{B}\right)$

If Q moves an amount $d\vec{l} = \vec{v}dt$

$$dW_{mag} = \vec{F}_{mag} \cdot d\vec{l} = Q\left(\vec{v} \times \vec{B}\right) \cdot \vec{v}dt = 0$$

since $\left(\vec{v} \times \vec{B}\right)$ Is perpendicular to $\left(\vec{v}\right)$

Magnetic forces may alter the direction in which a particle moves, but cannot speed it up or slow it down

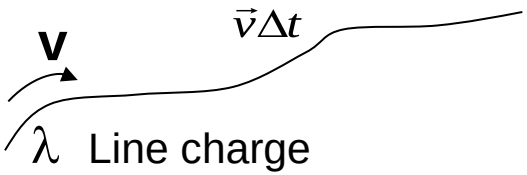
Currents

Charge per unit time passing a given point

Negatively charged electrons moves around-opposite to the direction of electric current

1 Ampere = 1 C / sec

Magnetostatics

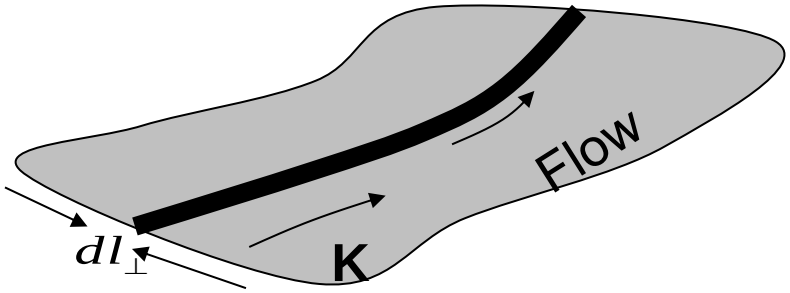


Current-carrying wire

$$I = \lambda v$$

Current is a vector

Charge-flow over a surface with a surface current density $\vec{K}$



The magnetic force due to a current is

$$\vec{F}_{mag} = \int \left(\vec{v} \times \vec{B}\right) dq$$

$$= \int \left(\vec{v} \times \vec{B}\right) \lambda dl = \int \left(\vec{I} \times \vec{B}\right) dl$$

Since I and dl point in the same direction

$$= \int I\left(d\vec{l} \times \vec{B}\right)$$

$$\vec{K} = \frac{dI}{dl_\perp}\hat{K} \qquad \vec{K} = \sigma\vec{v}$$

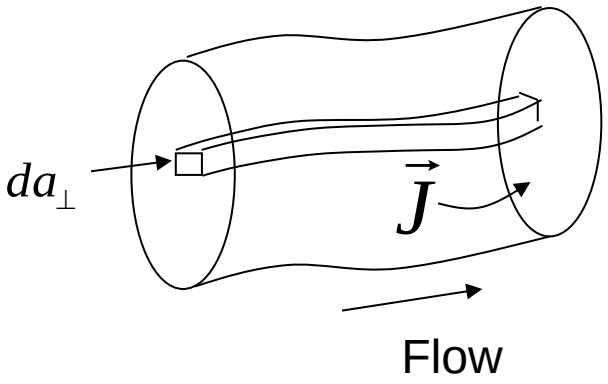
K is the current per unit width perpendicular to flow

$$\vec{F}_{mag} = \int \left(\vec{v} \times \vec{B}\right) \sigma da$$

$$= \int \left(\vec{K} \times \vec{B}\right) da$$

Magnetostatics

Volume current density $\vec{J}$



Flow of charge through a volume

$$\vec{J} = \frac{d\vec{I}}{da_\perp}$$

Current per unit area perpendicular to the flow

Also $\vec{J} = \rho \vec{v}$

$$\vec{F}_{mag} = \int \left(\vec{v} \times \vec{B} \right) \rho \, d\tau$$

$$= \int \left(\vec{J} \times \vec{B} \right) d\tau$$

Current crossing a surface S can be written as

$$I = \int_S J da_\perp = \int_S \vec{J} \cdot d\vec{a}$$

The total charge per unit time leaving a volume V is

$$\oint_S \vec{J} \cdot d\vec{a} = \int_V \left(\vec{\nabla} \cdot \vec{J}\right) d\tau$$

Because the charge is conserved, what comes out through the surface should come at the expense of the remaining inside

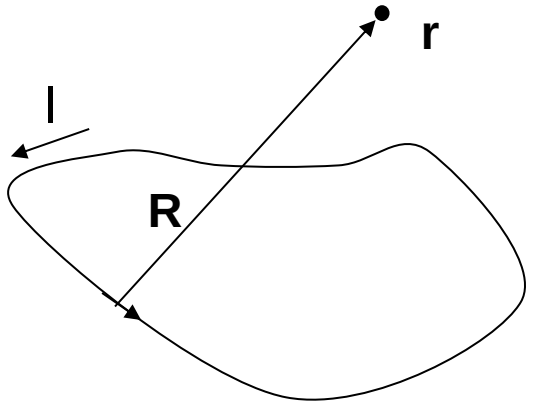
$$\int_V \left(\vec{\nabla} \cdot \vec{J}\right) d\tau = -\frac{d}{dt}\int_V \rho d\tau = -\int_V \frac{\partial \rho}{\partial t} d\tau$$

-ve sign implies decrease of charge inside the volume

True for any volume $\vec{\nabla} \cdot \vec{J} = -\frac{\partial \rho}{\partial t}$ **Continuity equation**

In magnetostatics $\frac{\partial \rho}{\partial t} = 0$ since we deal with steady currents

and hence $\vec{\nabla} \cdot \vec{J} = 0$



The Biot-Savart Law

Magnetic field of a steady line current is

$$\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} I \int \frac{d\vec{l} \times \hat{R}}{R^2}$$

Constant μ_0 is called permeability of free space

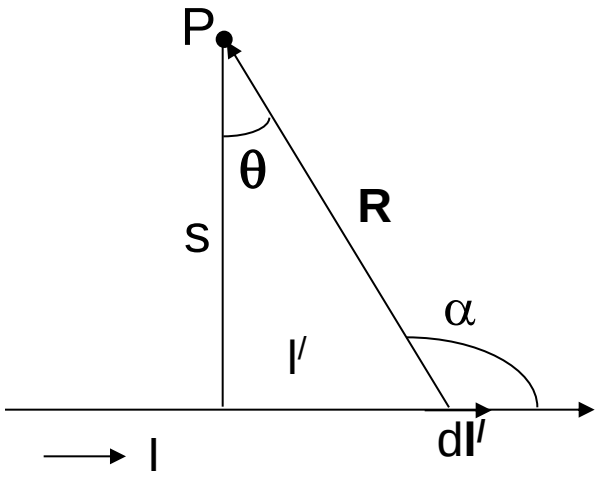
$\mu_0 = 4\pi \times 10^{-7}$ N / A^2

Integration is along the current path, in the direction of the flow, dl' is an element along the wire and R is the resultant vector

Biot-Savart law plays a role analogous to coulomb's law in electrostatics

Example

Magnetic field a distance s from a long straight wire carrying a steady current I



$d\vec{l}' \times \hat{R}$ points out of the page and has magnitude

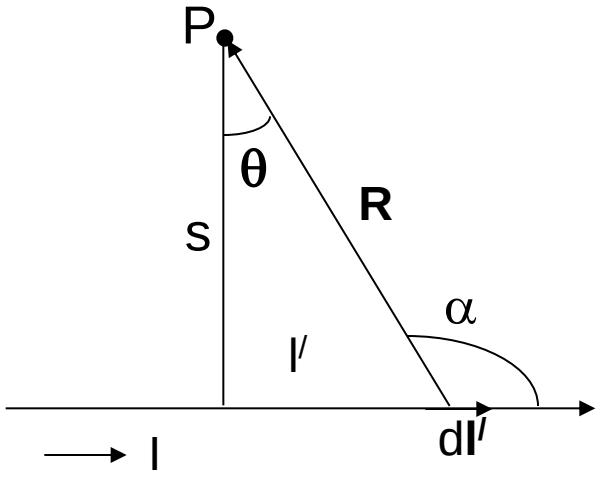
$$dl' \sin\alpha = dl' \cos\theta$$

$$l' = s \tan\theta$$

$$dl' = \frac{s}{\cos^2\theta} d\theta$$

$$s = R\cos\theta$$

$$\frac{1}{R^2} = \frac{\cos^2\theta}{s^2}$$



$$B = \frac{\mu_o I}{4\pi}\int_{\theta_1}^{\theta_2}\left(\frac{\cos^2\theta}{s^2}\right)\left(\frac{s}{\cos^2\theta}\right)\cos\theta d\theta$$

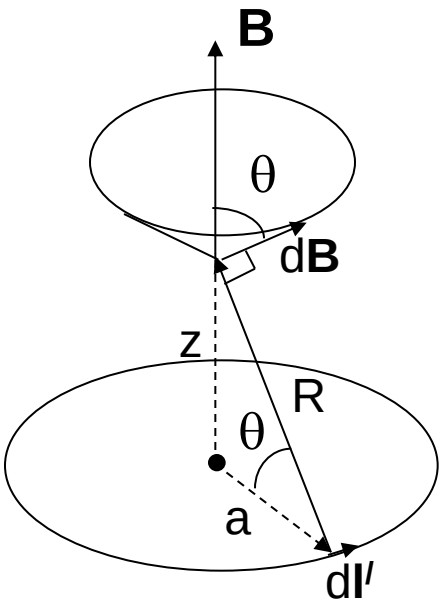
$$= \frac{\mu_0 I}{4\pi s}\left(\sin\theta_1 - \sin\theta_2\right)$$

In the case of infinite wire $\theta_1 = -\frac{\pi}{2}$ and $\theta_2 = \frac{\pi}{2}$

$$B = \frac{\mu_0 I}{2\pi s}$$

Example 2

Magnetic field at a distance z above the centre of a circular loop of radius a, which Carries a steady current I



Field d**B** is attributed to the line element d**l′**

Horizontal components cancels and vertical components combine to give

$$B(z) = \frac{\mu_0}{4\pi} I \int \frac{dl'}{R^2} \cos\theta$$

$\cos\theta$ and R^2 are constants

$$\int dl' = 2\pi a$$

$$B(z) = \frac{\mu_0 I}{4\pi}\left(\frac{\cos\theta}{R^2}\right) 2\pi a \quad = \frac{\mu_0 I}{2} \frac{a^2}{\left(a^2 + z^2\right)^{3/2}}$$

For surface and volume currents, Biot-Savart law becomes

$$\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi}\int \frac{\vec{K}(\vec{r}\,')\times\hat{R}}{R^2}da'$$

and

$$\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi}\int \frac{\vec{J}(\vec{r}\,')\times\hat{R}}{R^2}d\tau'$$

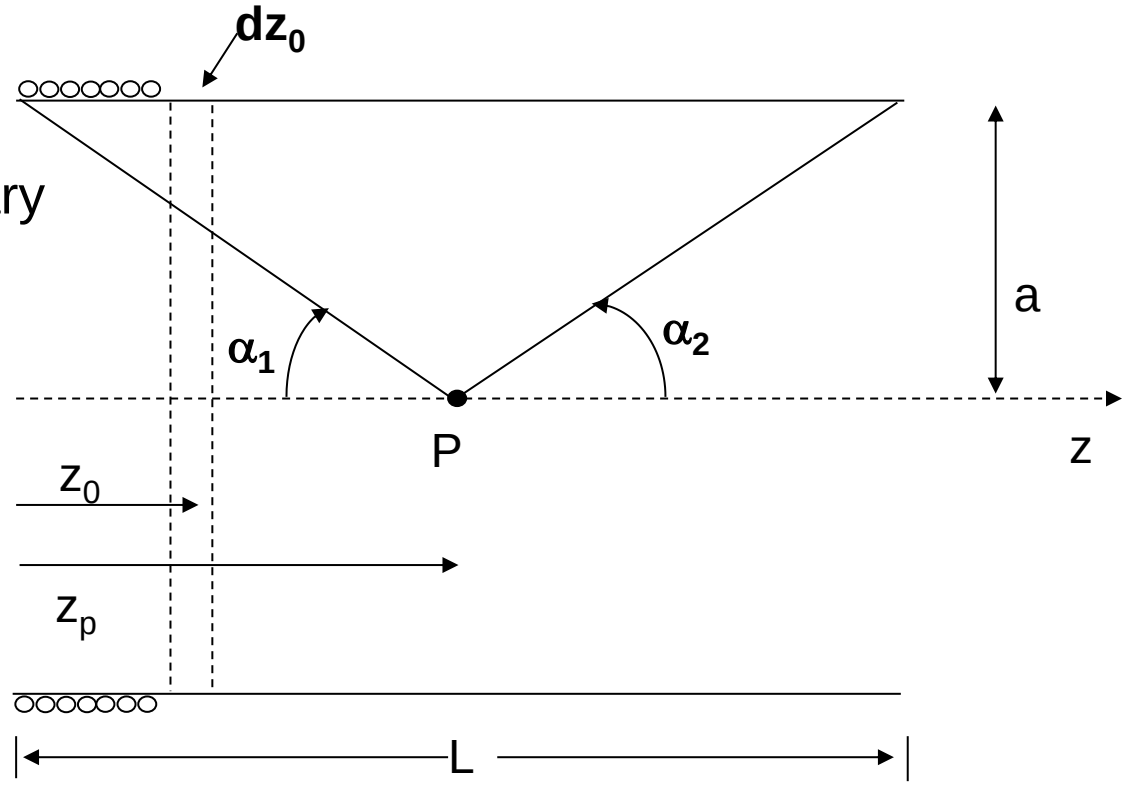
Axial induction of an ideal solenoid

N is the number of turns

Equivalent to N circular filamentary currents of radius a

Value of B at P – sum up the Contribution of each turn

Small portion dz_0 located at a distance z_0 from one end



$$n = N/L$$

no. of turns per unit length

$dN = ndz_0$ is the number of circular rings in the small section, each approximately $z = z_p - z_0$ from the field point P

$$dB = \frac{\mu_0 I a^2 n dz_0}{2\left[a^2 + \left(z_p - z_0\right)^2\right]^{3/2}}$$

Total value of B at P will be

$$B = \int_0^L \frac{\mu_0 I a^2 n dz_0}{2\left[a^2 + \left(z_p - z_0\right)^2\right]^{3/2}} = \frac{\mu_0 I n a^2}{2} \int_{-z_p}^{L-z_p} \frac{dz'}{\left(a^2 + z'^2\right)^{3/2}}$$

$$= \frac{\mu_0 I n}{2}\left\{\frac{\left(L - z_p\right)}{\left[a^2 + \left(L - z_p\right)^2\right]^{1/2}} + \frac{z_p}{\left(a^2 + z_p{}^2\right)^{1/2}}\right\}$$

used $z' = z_0 - z_p$

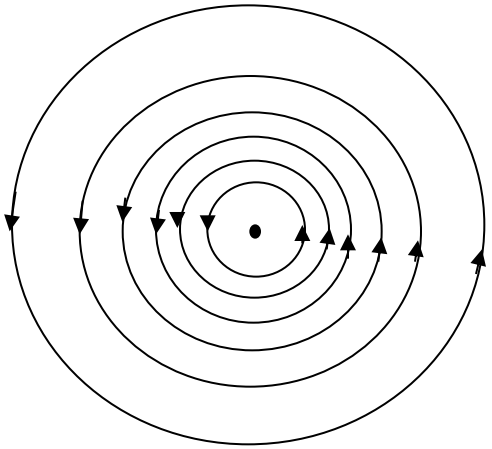
In terms of angles in the figure

$$B = \frac{1}{2}\mu_0 n I\left(\cos\alpha_1 + \cos\alpha_2\right)$$

For a solenoid of infinite length

$$B = \mu_0 n I$$

Divergence and curl of B



Integral of magnetic field of an infinite straight wire around a circular path of radius ‘s’ is

$$\oint \vec{B} \cdot d\vec{l} = \oint \frac{\mu_0 I}{2\pi s} dl = \frac{\mu_0 I}{2\pi s} \oint dl$$

$$= \mu_0 I$$

B is circular and ***I*** is perpendicular to the plane of the paper (coming outside)

We see, the answer is independent of the radius, since as B decreases, the circumference increases. In fact, it can be any loop which encloses the wire.

For instance, if we use cylindrical coordinates (s, ϕ, z) with current along z axis

$$\vec{B} = \frac{\mu_0 I}{2\pi s} \hat{e}_\phi$$

$$d\vec{l} = ds\hat{e}_s + sd\phi\hat{e}_\phi + dz\hat{e}_z$$

$$\oint \vec{B} \bullet d\vec{l} = \frac{\mu_0 I}{2\pi} \oint \frac{1}{s} s d\varphi = \frac{\mu_0 I}{2\pi} \oint d\phi = \mu_0 I$$

Bundle of straight wires

Each wire that passing through the loop contributes

$$\mu_0 I$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc}$$ where I_{enc} is the total current enclosed by the loop

If the flow of charge is represented by a volume current density $\vec{J}$

$$I_{enc} = \int \vec{J} \cdot d\vec{a}$$ Integral taken over the surface bounded by the loop

Applying stokes' theorem

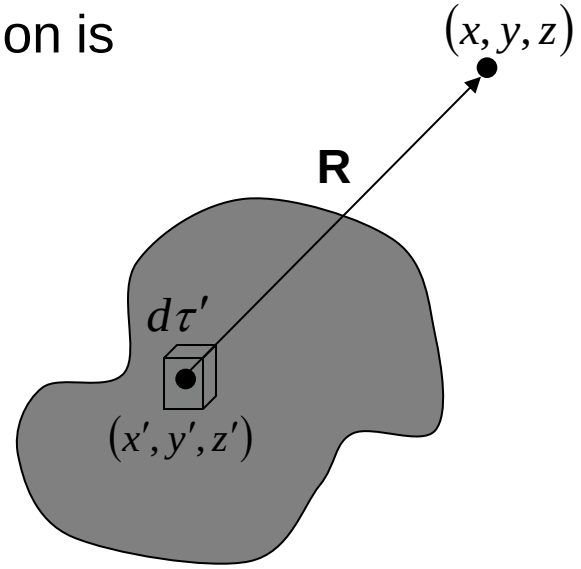
$$\int \left(\vec{\nabla} \times \vec{B}\right) \cdot d\vec{a} = \mu_0 \int \vec{J} \cdot d\vec{a}$$

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}$$ shows mag. field is rotational

Above derivation is restricted by the condition that we need infinitely straight line currents

Biot-Savart law for a volume current distribution is

$$\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi}\int \frac{\vec{J}(\vec{r}\,')\times \hat{R}}{R^2} d\tau'$$



$\vec{\mathbf{B}}$ is a function of (x, y, z)

$\vec{\mathbf{J}}$ is a function of (x', y', z')

$$\vec{R} = (x - x')\hat{e}_x + (y - y')\hat{e}_y + (z - z')\hat{e}_z$$

$$d\tau' = dx'dy'dz'$$

Integration is done over the primed coordinates
Divergence and curl is done over the unprimed coordinates

Applying divergence to the magnetic field B due to a volume charge distribution

$$\vec{\nabla}\cdot\vec{B}=\frac{\mu_0}{4\pi}\int\vec{\nabla}\cdot\left(\vec{J}\times\frac{\hat{R}}{R^2}\right)d\tau'$$

$$\vec{\nabla}\cdot\left(\vec{A}\times\vec{B}\right)=\vec{B}\cdot\left(\vec{\nabla}\times\vec{A}\right)-\vec{A}\cdot\left(\vec{\nabla}\times\vec{B}\right)$$

$$\vec{\nabla}\cdot\left(\vec{J}\times\frac{\hat{R}}{R^2}\right)=\frac{\hat{R}}{R^2}\cdot\left(\vec{\nabla}\times\vec{J}\right)-\vec{J}\cdot\left(\vec{\nabla}\times\frac{\hat{R}}{R^2}\right)$$

$\vec{\nabla}\times\vec{J}=0$ because $\vec{J}$ **does not depend on unprimed coordinates**

and $\vec{\nabla}\times\frac{\hat{R}}{R^2}=0$

$$\vec{\nabla}\cdot\vec{B}=0$$

Proves that mag. field is solenoidal

Applying curl to the magnetic field B due to a volume charge distribution

$$\vec{\nabla}\times\vec{B}=\frac{\mu_0}{4\pi}\int\vec{\nabla}\times\left(\vec{J}\times\frac{\hat{R}}{R^2}\right)d\tau'$$

This should be = and not +

$$\vec{\nabla}\times\left(\vec{A}\times\vec{B}\right)=\left(\vec{B}\cdot\vec{\nabla}\right)\vec{A}-\left(\vec{A}\cdot\vec{\nabla}\right)\vec{B}+\vec{A}(\vec{\nabla}\cdot\vec{B})-\vec{B}(\vec{\nabla}\cdot\vec{A})$$

$$\vec{\nabla}\times\left(\vec{J}\times\frac{\hat{R}}{R^2}\right)=\vec{J}\left(\vec{\nabla}\cdot\frac{\hat{R}}{R^2}\right)-\left(\vec{J}\cdot\vec{\nabla}\right)\frac{\hat{R}}{R^2}\longrightarrow\textbf{(a)}$$

The terms involving derivatives of $\vec{\mathbf{J}}$ is dropped since $\vec{\mathbf{J}}$ does not depend on (x,y,z)

The second term in (a) integrates to zero

$$\vec{\nabla}\cdot\frac{\hat{R}}{R^2}=4\pi\delta^3(\vec{R})$$

$$\vec{\nabla}\times\vec{B}=\frac{\mu_0}{4\pi}\int\vec{J}(\vec{r}\,')4\pi\delta^3(\vec{r}-\vec{r}\,')d\tau'=\mu_0\vec{J}(\vec{r})$$

Ampere's law

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}$$

is called Ampere's law (in differential form)

Using Stokes' theorem

$$\int \left(\vec{\nabla} \times \vec{B}\right) \cdot d\vec{a} = \oint \vec{B} \cdot d\vec{l} = \mu_0 \int \vec{J} \cdot d\vec{a}$$

$\int \vec{J} \cdot d\vec{a}$ is the total current passing through the surface-- I_{enc}

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc}$$

Ampere's law in integral form, also called Ampere's circuital law

For steady currents, the line integral of the magnetic induction vector B around a closed path is equal to μ_0 times the total current crossing any surface bounded by the line integral path.

Ampere's law has the same role as Gauss's law in electrostatics

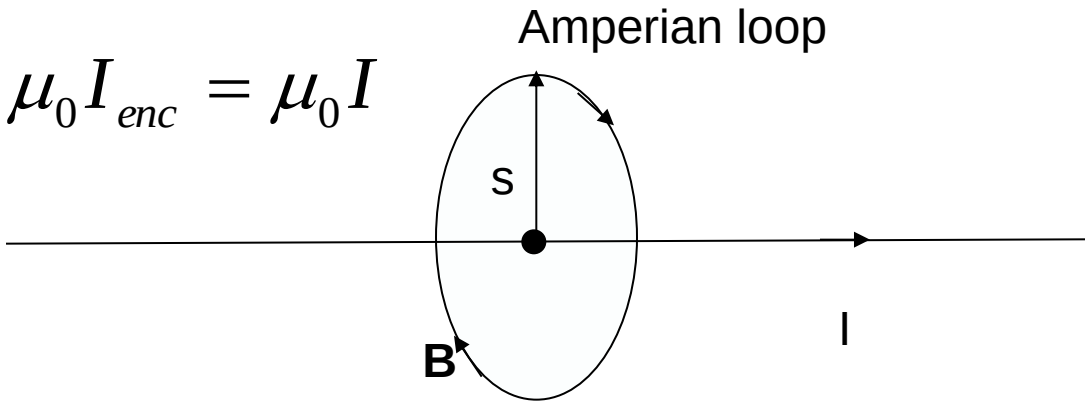
Example:

Find the magnetic field a distance s from a long straight wire carrying a steady current I

Direction of $\vec{B}$ is circumferential

By symmetry, the magnitude of $\vec{B}$ is constant around an amperian loop of radius s

$$\oint \vec{B} \cdot d\vec{l} = B \oint dl = B 2\pi s = \mu_0 I_{enc} = \mu_0 I$$



$$B = \frac{\mu_0 I}{2\pi s}$$

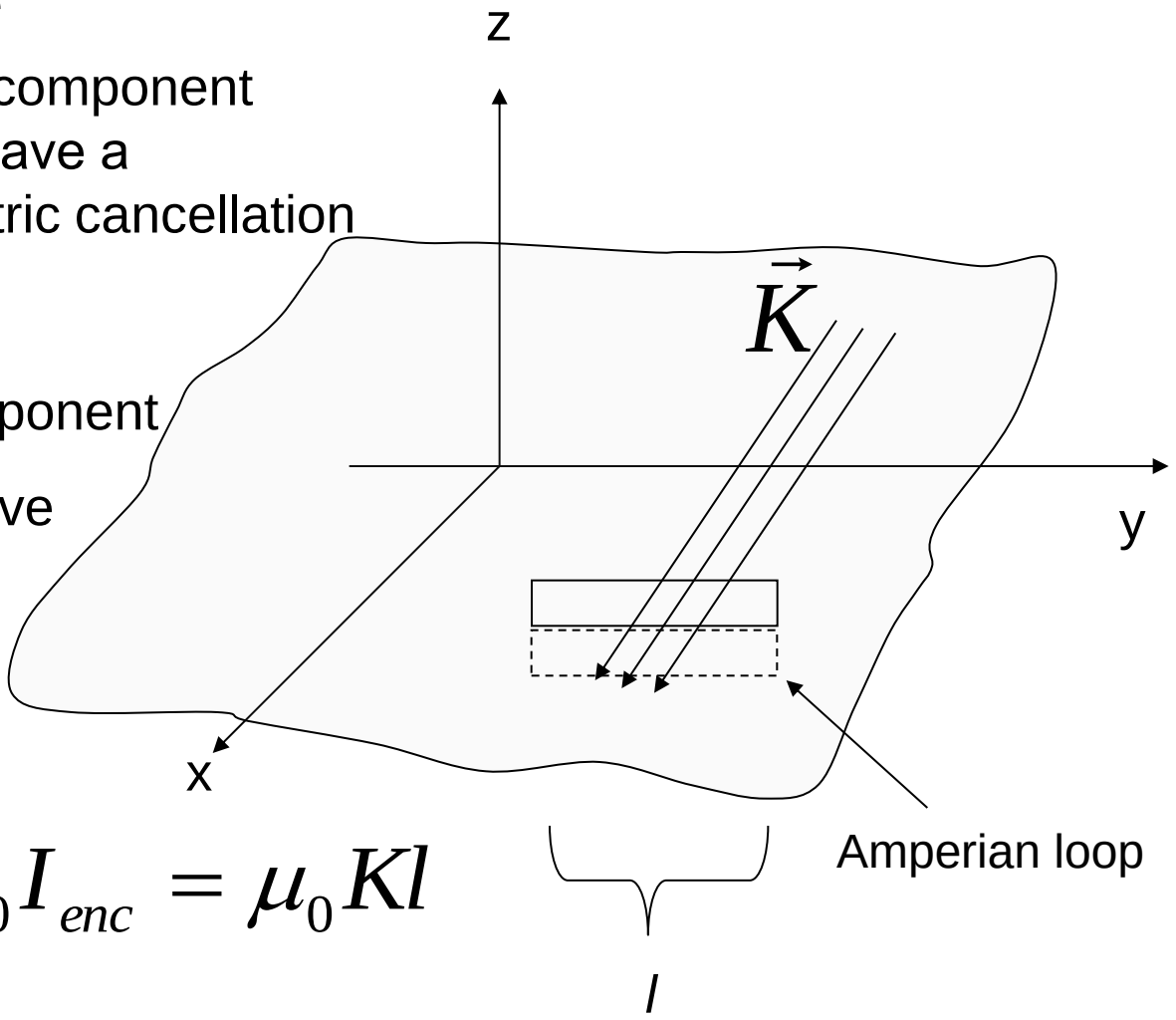
Example 2

Magnetic field of an infinite uniform surface current $\vec{K} = K\hat{e}_x$ flowing over the xy plane

The field cannot have an x- component since B $\perp$ K. It also cannot have a z-component due to symmetric cancellation from +y and –y elements

$\vec{\mathbf{B}}$ can only have a y-component

It points towards the left above the plane and towards the right in the plane below

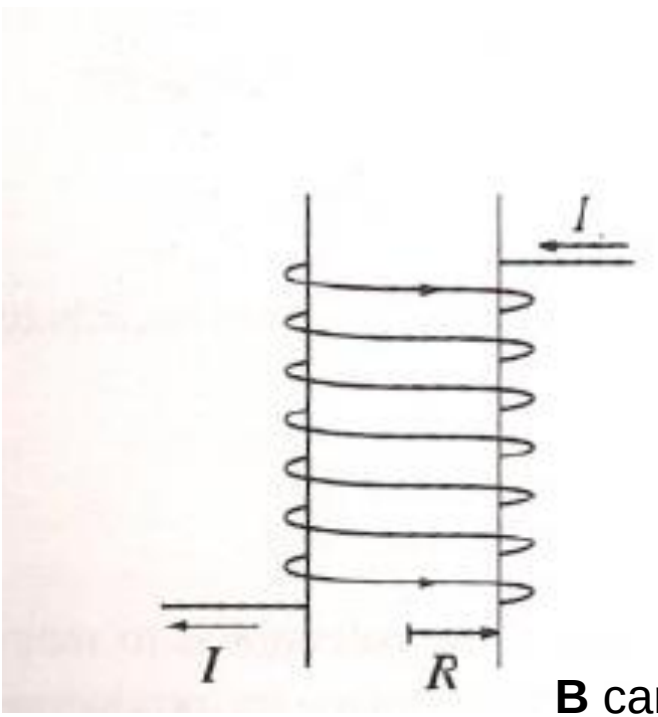


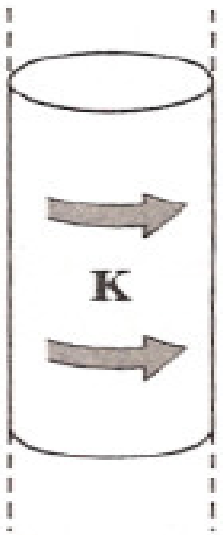
$$\oint \vec{B} \cdot d\vec{l} = 2Bl = \mu_0 I_{enc} = \mu_0 Kl$$

$$B = \left(\frac{\mu_0}{2} \right) K$$

Example 3

Magnetic field of a very long solenoid consisting of n closely wound turns per Unit length on a cylinder of radius R and carrying a steady current I

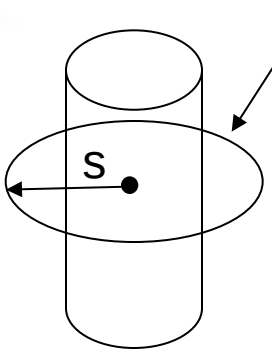




Surface current K is

$$K = nI$$

Amperian loop



B cannot have a radial component

It also will not have circumferential component B_ϕ , why?

If it is non-zero, B_ϕ would be constant around the amperian loop concentric to the solenoid and hence

$$\oint \vec{B} \cdot d\vec{l} = B_\phi (2\pi s) = \mu_0 I_{enc} = 0$$

Since the loop encloses no current

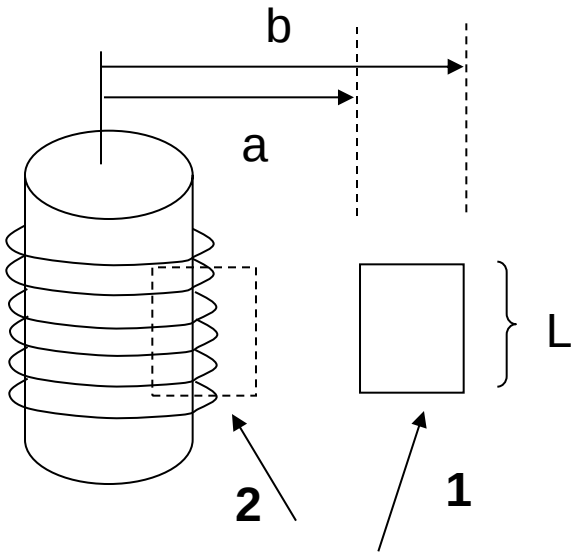
So the magnetic field of an infinite solenoid runs parallel to the axis

It points upward inside the solenoid and downward outside

Loop1

$$\oint \vec{B}\cdot d\vec{l} = [B(a)-B(b)]L = \mu_0 I_{enc} = 0$$

$$B(a) = B(b)$$



Amperian loops

Field outside does not depend on the distance from the axis!!

But field will go to zero at large *s*

Hence it will be zero everywhere!!!

Loop 2

$$\oint \vec{B}\cdot d\vec{l} = BL = \mu_0 I_{enc} = \mu_0 nIL$$

$$\vec{B} = \begin{cases} \mu_0 nI\hat{z} & \text{Inside the solenoid} \\ 0 & \text{Outside the solenoid} \end{cases}$$

Like Gauss' law, Ampere's law is always true (for steady currents)

However it is mainly in symmetry cases where B can be pulled out of the integral